Zero as a Dimension Switch: A Conceptual Framework for Reinterpreting the Number Line

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Abstract

In the conventional real number line, zero is positioned at the center between positive and negative values. This paper proposes a reinterpretation of zero as a *dimension switch* — a non-numeric boundary between two distinct spaces. The key feature is the **Reflection Rule** governing operations that cross the zero boundary, ensuring magnitude conservation while switching the dimensional state.

1 Introduction

In standard arithmetic, zero acts as the additive identity and lies at the center of symmetry in the real number line. This conceptual framework explores treating zero not as a number but as a transition point between two dimensions, allowing for a fresh approach to quantities that naturally conserve magnitude while possessing opposite directions.

2 Motivation

The primary motivation for this reinterpretation is to develop a structure that handles opposition (e.g., $a+(-a)$) as a shift in state rather than a mere cancellation to an additive identity.

1. **Directional Conservation:** Modeling physical phenomena (e.g., mass, energy) where magnitude cannot be negative, but direction/state can be reversed (anti-matter analogy).
2. **Algebraic Clarity:** Replacing the traditional symmetry of the real line with a **V-shaped reflection model**, where all numbers are positive in their respective space.

3 Definition of Spaces

We define:

- $\mathbb{R}^+$: the *positive space*, containing elements $x = (m, +1)$ where $m \in \mathbb{R}_{\geq 0}$.

- $\mathbb{R}^-$: the *mirror space*, containing elements $x^- = (m, -1)$ where $m \in \mathbb{R}_{\geq 0}$.
- 0: the non-numeric *switch point* or *boundary*, where the magnitude $m = 0$.

Elements of $\mathbb{R}^-$ are symbolic, corresponding to the magnitude in the mirror dimension.

The overall structure can be conceptually viewed as:

$$\mathbb{S} = (\mathbb{R}_{\geq 0} \times \{+1, -1\}) \,/\, \sim$$

where the equivalence relation $\sim$ identifies $(0, +1)$ and $(0, -1)$ with the single boundary point 0.

4 Transformation Across Zero

We introduce a *space transformation* (Dimension Switch operator):

$$T : \mathbb{R}^+ \to \mathbb{R}^-, \quad T((m, +1)) = (m, -1) = m^-$$

and its inverse

$$T^{-1} : \mathbb{R}^- \to \mathbb{R}^+, \quad T^{-1}((m, -1)) = (m, +1) = m$$

These transformations formalize the dimensional change upon crossing the boundary.

5 Arithmetic Axioms (The Reflection Rule)

We define the primary operation $\ominus$ (generalized subtraction) for elements $x = (m_x, s_x)$ and $y = (m_y, s_y)$, focusing on the case where $x \in \mathbb{R}^+$ and y is the inverse operation (subtraction of magnitude m_y towards the origin).

5.1 Subtraction Crossing the Boundary

If we subtract a magnitude m_y from a positive element m_x:

$$m_x \ominus m_y = \begin{cases} (m_x - m_y, +1) & \text{if } m_x \geq m_y \quad \text{(Stays in } \mathbb{R}^+\text{)} \\ (m_y - m_x, -1) & \text{if } m_x < m_y \quad \text{(Reflects to } \mathbb{R}^-\text{)} \end{cases}$$

This axiom models the "bounce" or reflection off the 0 boundary, ensuring that the net magnitude is conserved, but the dimensional state flips.

5.2 Continuity and Limits

In this framework, the **magnitude** of a quantity is continuous across 0:

$$\lim_{t \to 0} |f(t)| = 0$$

The discontinuity is confined only to the discrete state variable $s \in \{+1, -1\}$, not the physical quantity m. This addresses the ambiguity of continuity present in previous interpretations.

6 Potential Applications

Potential mathematical and physical applications include:

- Modeling systems where the absolute value (energy/mass) is conserved during a directional reversal (e.g., anti-matter creation).
- Providing an alternative basis for vector spaces where the directional axis is explicitly separated from the magnitude.
- Exploring extensions of algebra where the additive identity is replaced by a "state annihilation" or "boundary condition."

7 Future Work and Open Questions

Future developments may include:

- Rigorous algebraic formalization of multiplication and addition involving elements from both $\mathbb{R}^+$ and $\mathbb{R}^-$.
- Geometric representation using animation tools such as Manim.
- Analyzing the compatibility of this structure with existing algebraic fields.

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